Digi-Key Purchase Request

Duotao Gao | EGR 304 | Team 103 GripRx | October 29, 2026

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UI board components for breadboard prototyping and PCB assembly. Requested quantities include planned spares. Already reported available parts are not ordered again.

Materials subtotal: $37.59. Course allowance: $60 per member. PRLTA components/PCBs, development kits, free and out-of-pocket components are excluded. Tax/shipping are unquoted; please confirm their budget treatment.

Qty	Digi-Key item #	Part / MPN	Unit USD	Line USD
2	NHD-0216K1Z-FSW-FBW-L-ND	16 x 2 character LCD module / NHD-0216K1Z-FSW-FBW-L	$8.11	$16.22
4	1189-16YXJ220MCR6.3X11-ND	Electrolytic capacitor, 220 uF 16 V / 16YXJ220MCR6.3X11	$0.45	$1.80
4	445-FA28X7R1H334KRU00-ND	Nonpolar ceramic capacitor, 0.33 uF 50 V / FA28X7R1H334KRU00	$0.37	$1.48
4	445-180206-ND	Nonpolar ceramic capacitor, 1 uF 50 V (16 V minimum) / FG28X5R1H105KRT00	$0.31	$1.24
4	200KXBK-ND	Resistor, 200k / MFR-25FBF52-200K	$0.11	$0.44
4	1801-1N5822CT-ND	Schottky diode, reverse-polarity protection / 1N5822	$0.71	$2.84
3	732-5395-ND	Shrouded ribbon header, 2 x 4 / 61200821621	$1.03	$3.09
3	ED2635-ND	Speaker screw terminal, 2 positions / OSTTE020161	$0.67	$2.01
3	2223-CMS-3652-28L200-ND	Speaker, 8 ohm, 2 W / CMS-3652-28L200	$2.17	$6.51
2	ATS2127-ND	TO-220 heatsink / ATS-PCBT1085	$0.68	$1.36
6	S9001-ND	Two-pin jumper shunt / SPC02SYAN	$0.10	$0.60

Materials subtotal $37.59 | Tax/shipping TBD | Grand total TBD

Approval: _____ Date: _____

Product links and selection notes

J4 - NHD-0216K1Z-FSW-FBW-L

https://www.digikey.com/en/products/detail/newhaven-display-intl/NHD-0216K1Z-FSW-FBW-L/1701185

Two LCDs: one working display can move from breadboard to socketed PCB; the second is a spare. The optional third is omitted to control cost. Course module approval still requires confirmation.

C14 - 16YXJ220MCR6.3X11

https://www.digikey.com/en/products/detail/rubycon/16YXJ220MCR6-3X11/5429971

220 uF 16 V polarized. Course-list row 41 has no order confirmation, so it is purchased. Active Rubycon part, $0.45 each, 2654 in stock when checked. Verify positive pad and clearance.

C2 - FA28X7R1H334KRU00

https://www.digikey.com/en/products/detail/tdk/FA28X7R1H334KRU00/7384656

0.33 uF 50 V X7R nonpolar. Course-list row 104 is under Not Necessary this Fall and is not treated as issued stock. Selected active part: $0.37 each, 446 in stock when checked.

C9 - FG28X5R1H105KRT00

https://www.digikey.com/en/products/detail/tdk/FG28X5R1H105KRT00/7384818

1 uF 50 V nonpolar ceramic; exceeds original 16 V rating. X5R capacitance varies with bias; verify audio response. Match radial footprint.

R13 - MFR-25FBF52-200K

https://www.digikey.com/en/products/detail/yageo/MFR-25FBF52-200K/13547

Four per supplied reference. Existing resistor footprint nominal 6.3 mm body and 10.16 mm lead spacing; check hole diameter.

D3 - 1N5822

https://www.digikey.com/en/products/detail/taiwan-semiconductor-corporation/1N5822/7366597

40 V 3 A Schottky. Select Cut Tape, not full reel. Existing kit rectifier is not assumed equivalent.

J1 - 61200821621

https://www.digikey.com/en/products/detail/w%C3%BCrth-elektronik/61200821621/4846916

Match numbered pins, key orientation and the teammate cable before PCB.

J5 - OSTTE020161

https://www.digikey.com/en/products/detail/on-shore-technology-inc/OSTTE020161/614584

Speaker connects by wires. Verify pad pitch and terminal pin mapping.

External speaker via J5 - CMS-3652-28L200

https://www.digikey.com/en/products/detail/same-sky-formerly-cui-devices/CMS-3652-28L200/29778294

Three per supplied reference. Keep current amplifier design; actual loudness still requires testing.

U1 heatsink - ATS-PCBT1085

https://www.digikey.com/en/products/detail/advanced-thermal-solutions-inc/ATS-PCBT1085/5030505

Two selected within reference range 2-3. Clip attachment avoids separate screws. Verify temperature with actual regulator dissipation and enclosure.

JP1, JP2 shunts - SPC02SYAN

https://www.digikey.com/en/products/detail/sullins-connector-solutions/SPC02SYAN/76375

Six per supplied reference. Quantity 10 price is 0.029 each (0.29 total), cheaper than six at 0.10; retained six to follow reference quantities.

The 11-item bulk-upload CSV is included to facilitate cart entry. A shared-cart link has not been created; the assignment requests it when possible.